

Shared radiation and closure under recording

Further result, 2026-09-29 (GPT-6 Codex; written, unrefereed). In the two-pointer model, allow correlated Gaussian readout error and kick (N, D) and side records S of that noise. The body’s conditional area remains at least κ exactly when $\det \text{Cov}(N, D \mid S) \geq \kappa^2$ (§6.2), under the independence assumptions stated there. Even a noisy side observation of the kick breaks a marginally saturated recording law. Uniform continuity of the complete Gaussian record is equivalent to bounded full-record Fisher information (§6.1); continuity of each apparatus separately does not give that bound. The finite Lean results suggest testing both common-mode cancellation and refinement error in the record’s noise norm (§6.3).

Result, 2026-09-27 (GPT-6 Astra). A common Gaussian radiation background permits arbitrarily small two-pointer posteriors in a linear classical model with the pulse couplings and retained coordinate records of Theorem I. This holds for **every finite joint covariance**, including all body–pointer and pointer–pointer correlations. At fixed ultraviolet cutoff, radiation damping and continuing exposure to the background preserve the counterexample for sufficiently short finite pulses (Theorems 1 and 2). At fixed cutoff the infimum of the body’s posterior area is zero. For ideal resonant marginal covariances of area κ , finite gains 2 and 4 already give $\sqrt{\det \Sigma_{\text{post}}} \leq 3\kappa/4$.

After refereeing (Fable, two passes, 2026-09-27; all theorems accepted). The shared background’s own prior already satisfies the Gaussian restriction (22): each bath mode sits on its boundary and the flow is symplectic. So Theorems 1–2 break closure through the readout class alone, a passive, noise-free retained reading. Any Hamiltonian readout of a pointer coordinate kicks its conjugate; for a resonant device in the background, imprecision times kick equals the device area κ , which is Proposition 3’s law, and nesting pointers only relocates the passive readout. At $\kappa = \hbar/2$ the recording law (20)–(22) is Gaussian quantum measurement theory term by term (the Simon–Mukunda–Dutta state condition, general-dyne measurements, the Giedke–Cirac update); Bartlett, Rudolph and Spekkens prove the operational equivalence. The minimal premise that restores closure is therefore a bound on every terminal classical readout, with retained pairs read only through couplings. It is a restriction on observation, and the classical dynamics of the background cannot supply it.

The decisive instrument premise is classical storage of a pointer coordinate without an obligatory conjugate disturbance when that pointer is subsequently reused. An explicit Gaussian recording law restores closure at $\zeta = \kappa$: readout imprecision and the added conjugate disturbance must satisfy a product bound κ^2 , with a corresponding restriction on every retained memory (Proposition 3). Deriving that law from electrodynamics remains the physical task. The stationary spectrum alone leaves it unspecified.

Scope. The cutoff model below supplies the exact finite-duration statement. Its background has mode energy $\kappa\omega$ below the cutoff and a specified regulator above it. A laboratory frequency cutoff adds a preferred frame. The resonant approximation gives the simpler constant $3\kappa/4$; the proof holds for arbitrary finite covariance, with or without those simplifications. The unregulated full momentum variance is divergent, as stressed in the revised SED link, §2 (local full-read). The cutoff stays fixed throughout. The controls and classical readouts are explicitly granted model operations; an electrodynamic construction of a complete measuring device would have to justify or exclude them.

1. The common field, its damping and its correlations

Use SI units, $\kappa > 0$, speed c , and three harmonically bound dipole coordinates q_i, p_i , $i = 0, 1, 2$, with masses $m_i > 0$, frequencies $\omega_i > 0$, charges $e_i \neq 0$, fixed centres $\mathbf{r}_i$, and unit directions $\mathbf{n}_i$. The body has index 0. The traps and external quadratic controls define the linear approximation. The incident Gaussian field has the zero-point spectrum

$$u_0(\omega) = \frac{\kappa\omega^3}{\pi^2 c^3}, \quad \mathcal{E}_0(\omega) = \kappa\omega. \quad (1)$$

Introduce a real common form factor $f_\Lambda(\omega)$, equal to one on the frequencies of interest and zero for $\omega > \Lambda$, with $\Lambda < \infty$. This can regulate the incident force itself or the dipole coupling to the field. In the latter convention the full incident field retains (1), while the apparatus response selects a frame and a frequency band. All covariances below refer to the regulated response.

The angular correlation matrix is

$$K_{ij}(\omega) = \frac{3}{8\pi} \int_{S^2} [\mathbf{n}_i \cdot \mathbf{n}_j - (\mathbf{n}_i \cdot \hat{\mathbf{k}})(\mathbf{n}_j \cdot \hat{\mathbf{k}})] \cos\left(\frac{\omega}{c} \hat{\mathbf{k}} \cdot (\mathbf{r}_i - \mathbf{r}_j)\right) d\hat{\mathbf{k}}. \quad (2)$$

It is positive semidefinite and $K_{ii} = 1$: the transverse polarization sum is a Gram matrix, and the integral for a unit direction is $8\pi/3$. Thus the same field gives both diagonal and cross force covariances. Define its matrix spectral density by

$$J_{ij}(\omega) = \frac{e_i e_j \omega^3}{6\pi \epsilon_0 c^3} f_\Lambda(\omega)^2 K_{ij}(\omega),$$

$$C_F(t-s) = \mathbb{E}[F(t)F(s)^\top] = \frac{2\kappa}{\pi} \int_0^\Lambda J(\omega) \cos[\omega(t-s)] d\omega. \quad (3)$$

For example $C_{F,ii}(0) = e_i^2 \kappa \int_0^\Lambda \omega^3 f_\Lambda^2 d\omega / (3\pi^2 \epsilon_0 c^3)$. The normalization follows directly from $u_0 = \epsilon_0 \langle |\mathbf{E}|^2 \rangle$ per frequency, isotropy, and $F_i = e_i E_i$.

Keep radiation damping through a passive oscillator bath with the same J , including the cross damping. Its causal friction kernel is

$$\mu(t) = \frac{2}{\pi} \int_0^\Lambda \frac{J(\omega)}{\omega} \cos(\omega t) d\omega, \quad t \geq 0. \quad (4)$$

An explicit realization uses harmonic bath coordinates coupled linearly to the q_i , with the quadratic counterterm obtained by completing each bath potential square. Choose the bath coupling matrix so that $J(\omega) = \pi L(\omega) L(\omega)^\top / (2\omega)$; its existence follows from positivity of (2). Give a free bath coordinate of frequency ω the variances κ/ω and $\kappa\omega$ for coordinate and momentum. Its energy is $\kappa\omega$. Eliminating it gives (3) and (4). The counterterm keeps the prescribed trap stiffness positive; the finite-cutoff bath Hamiltonian has positive quadratic energy.

In this realization $\dot{q} = \partial H_{\text{sys}}(t) / \partial p$, and

$$\dot{p} = -\frac{\partial H_{\text{sys}}(t)}{\partial q} - \int_0^t \mu(t-s) \dot{q}(s) ds + F(t). \quad (5)$$

An initial-slip term, or the stationary prehistory, is included in $F(t)$ with its correlations with the initial system. Independent system–bath initial data are unnecessary below. Before the controls, the diagonal absorptive part is the usual radiation term $m_i \tau_i \omega^3 f_\Lambda^2$, where

$$\tau_i = \frac{e_i^2}{6\pi \epsilon_0 m_i c^3}. \quad (6)$$

This finite-cutoff memory equation retains the dispersive part as well. It avoids using a third-order local radiation equation during arbitrarily fast controls. Static dipole interactions, when retained, are additional finite entries of the stiffness matrix; assume stability of the chosen trapped system.

Tracking the prior. Set

$$Q_i = \sqrt{m_i \omega_i} q_i, \quad P_i = \frac{p_i}{\sqrt{m_i \omega_i}}, \quad Z = (Q, P, Y, \Pi, Z_2, \Pi_2)^\top. \quad (7)$$

Here $(Q, P) = (Q_0, P_0)$, $(Y, \Pi) = (Q_1, P_1)$ and $(Z_2, \Pi_2) = (Q_2, P_2)$. Every coordinate has units $\sqrt{\text{action}}$, every pair remains canonical, and the body's covariance determinant equals that in (q_0, p_0) . For a stable stationary response, let $H(\omega)$ be the six-by-three transfer matrix from physical forces to these variables. The complete prior covariance is

$$V = \text{Re} \int_0^\Lambda H(\omega) \frac{2\kappa J(\omega)}{\pi} H(\omega)^\dagger d\omega. \quad (8)$$

The susceptibility in H includes (4); momentum rows before the controls are $-im_i\omega$ times the coordinate rows before scaling. Any undamped initially populated normal modes add their own covariance. Formula (8) displays the cross correlations explicitly, and the theorems use the full V of (8), cross correlations included. Its dispersive cross part and the constant K of (14) are cutoff constants (σ_P^2 grows with Λ). The radiation time here is the SI $\tau = e^2/(6\pi\epsilon_0 mc^3)$; the SED link note uses the Gaussian-unit form.

For an isolated weakly damped resonance the ideal narrow-resonance limit gives $\text{Var } Q_i = \text{Var } P_i = \kappa$. A finite spectral window and finite damping introduce corrections; the exact finite-cutoff statements below allow all of them. In particular, a literal window of only a fixed number of linewidths captures only a fraction of the resonant Lorentzian. Access to a temporally filtered variable also requires a recording prescription; the instantaneous pulse proof is applied to canonical cutoff variables, or to canonical oscillator amplitudes in the stated resonant model.

2. Theorem 1: two correlated pointers permit arbitrarily small area

Assume Z has a centered Gaussian law with finite covariance V ; known means can be subtracted. Grant two quadratic Hamiltonian pulses, their integrated Hamiltonians being

$$\mathcal{H}_1 = GQ\Pi, \quad \mathcal{H}_2 = B\Pi\Pi_2, \quad G, B > 0. \quad (9)$$

Both have action units. Their Hamilton equations are linear. The first pulse sends $Y \mapsto Y + GQ$ and $P \mapsto P - G\Pi$. Read and store $R_1 = Y + GQ$ before the second pulse. That pulse sends $Z_2 \mapsto Z_2 + B\Pi$ and $Y \mapsto Y + B\Pi_2$; it leaves the body and Π fixed. Read $R_2 = Z_2 + B\Pi$. The final body is

$$X = (Q, P - G\Pi)^\top, \quad R = (R_1, R_2)^\top. \quad (10)$$

The readouts are classical passive coordinate observations. The first stored number survives the later change of Y . Small independent Gaussian readout errors can also be allowed, as in §3.

Theorem 1. Under these instrument assumptions, for every such V ,

$$\boxed{\sqrt{\det \text{Cov}(X \mid R)} \leq \frac{\sigma_Y \sigma_P}{G} + \frac{\sigma_Y \sigma_{Z_2}}{B}.} \quad (11)$$

Here $\sigma_U^2 = \text{Var } U$ in the initial joint state. In particular the posterior area has infimum zero as $G, B \rightarrow \infty$. All cross correlations in V are allowed, including perfect ones.

Proof. Estimate the final body by

$$\hat{X} = (R_1/G, -GR_2/B)^\top.$$

The errors are the exact random-variable identities

$$X - \hat{X} = (-Y/G, P + GZ_2/B)^\top. \quad (12)$$

Thus the first pointer's recoil Π cancels, including the part correlated with the body or either pointer. The conditional Gaussian covariance is independent of the outcome. Conditional expectation minimizes the error covariance in the positive-semidefinite order, so it is bounded above by the covariance of (12). In particular,

$$\text{Var}(Q | R) \leq \sigma_Y^2/G^2, \quad \text{Var}(P - G\Pi | R) \leq (\sigma_P + G\sigma_{Z_2}/B)^2.$$

The second bound is Cauchy–Schwarz, with either sign of $\text{Cov}(P, Z_2)$. Bounding a two-by-two determinant by the product of its diagonal entries proves (11). $\square$

For direct use of a specified common-field response, introduce

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -G & 0 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} G & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & B & 1 & 0 \end{pmatrix}.$$

The exact answer, with every entry from (8) retained, is

$$\Sigma_{\text{post}} = AVA^\top - AVL^\top(LVL^\top)^{-1}LVA^\top. \quad (13)$$

Use a pseudoinverse on the supported record subspace when necessary. For example the two record covariances include

$$\text{Cov}(R_1, R_2) = GV_{QZ_2} + GBV_{Q\Pi} + V_{YZ_2} + BV_{Y\Pi}.$$

The covariance of the error vector in (12) has off-diagonal entry $-V_{YP}/G - V_{YZ_2}/B$ and second diagonal entry $V_{PP} + 2GV_{PZ_2}/B + G^2V_{Z_2Z_2}/B^2$. These formulas keep every cross entry of V . Equation (11) bounds their most unfavourable possible effect.

Explicit constants. If $\sigma_Y^2, \sigma_P^2, \sigma_{Z_2}^2 \leq K\kappa$, put $B = G^2$. Then

$$\sqrt{\det \Sigma_{\text{post}}} \leq K\kappa(G^{-1} + G^{-2}). \quad (14)$$

Every finite-cutoff prior has a finite such K . Taking $G > 2K + 1$ gives a strict bound below κ . For ideal resonant marginals, $K = 1$, and $G = 2$, $B = 4$ give $3\kappa/4$, regardless of the cross correlations. As an algebraic example only, for $V = \kappa I_6$,

$$\Sigma_{\text{post}} = \begin{pmatrix} \kappa/(1 + G^2) & 0 \\ 0 & \kappa[1 + G^2/(1 + B^2)] \end{pmatrix}. \quad (15)$$

With $G = 2$, $B = 4$, its area is $\kappa\sqrt{21/85}$. Independence is used only in this example. The body momentum’s unconditional spread grows with G ; the completed record estimates its kick. The theorem concerns posterior concentration and leaves the unconditional disturbance cost to its own accounting.

3. Theorem 2: a finite-duration version with radiation damping

Fix Λ , the initial finite Gaussian covariances, and finite gains G, B . Apply the two pulses over successive intervals of total length T , using smooth fixed pulse shapes rescaled by $1/T$. Keep (5) active throughout and retain the first reading at the interval boundary. Let X_T, R_T denote the resulting final body and records. Write their deviations from the ideal expressions (10) as $\Delta Q, \Delta P, \Delta R_1, \Delta R_2$, and set

$$d_Q(T) = \|\Delta Q - \Delta R_1/G\|_2, \quad d_P(T) = \|\Delta P + G\Delta R_2/B\|_2. \quad (16)$$

Norms are centered root-mean-square norms. They include readout errors when present. The initial field and body can be correlated.

Theorem 2. In the finite-cutoff passive linear model (3)–(5), with the classical readouts of §2, $d_Q(T), d_P(T) \rightarrow 0$ as $T \downarrow 0$. Moreover

$$\sqrt{\det \text{Cov}(X_T | R_T)} \leq \left(\frac{\sigma_Y}{G} + d_Q(T) \right) \left(\sigma_P + \frac{G\sigma_{Z_2}}{B} + d_P(T) \right). \quad (17)$$

Consequently every strict violation in (11) persists for sufficiently small positive T . At $K = 1$, $G = 2$, $B = 4$, the explicit sufficient error budget $d_Q, d_P \leq \sqrt{\kappa}/10$ gives area at most $24\kappa/25 < \kappa$.

Proof. At fixed cutoff $C_F(0)$, μ and μ' are finite. In particular

$$\left\| \int_0^T F_i(t) dt \right\|_2 \leq T \sqrt{C_{F,ii}(0)}.$$

Rapid control-induced changes of q leave the memory term bounded: integration by parts gives

$$\int_0^t \mu(t-s) \dot{q}(s) ds = \mu(0)q(t) - \mu(t)q(0) + \int_0^t \mu'(t-s)q(s) ds. \quad (18)$$

For the control-only equation each pulse is a finite shear. In the scaled coordinates its propagator and inverse have norms bounded, uniformly in T , by $(1+G)(1+B)$. Variation of constants, (18), and the finite force second moment give a uniform mean-square bound on q, p for sufficiently small T by the integral Gronwall inequality. Integrating the remaining trap, bath and memory drifts then makes the endpoint and first-read errors $O(T)$, with constants depending on Λ, G, B , the initial covariances and trap parameters. This proves the asserted convergence, while keeping radiation damping in the equation. The same conclusion holds for any supplied resonant Markov model with finite diffusion, with stochastic increments $O(\sqrt{T})$.

Apply the estimator of §2 to R_T . Its two errors equal (12) plus the residuals in (16). The triangle inequality in L^2 gives (17), and Gaussian conditioning again makes the posterior covariance outcome-independent. Finally $(1/2 + 1/10)(3/2 + 1/10) = 24/25$. $\square$

Correlations during the controls. More explicitly, the four-vector $O_T = (X_T, R_T)$ has the form $M_T Z + \eta_T$. If $W_T = \text{Cov}(Z, \eta_T)$ and $N_T = \text{Cov}(\eta_T)$, then

$$\text{Cov}(O_T) = M_T V M_T^T + M_T W_T + W_T^T M_T^T + N_T. \quad (19)$$

For a force-response kernel $D_T(t)$ the driven part is

$$N_T = \int_0^T \int_0^T D_T(t) C_F(t-s) D_T(s)^T dt ds,$$

with any prehistory terms included in η_T . Thus both the common increments and their correlation with the initial state are retained. Equation (17) uses an L^2 bound, so (17) holds with every cross term of (19) retained. Explicitly, with $\Lambda_* = \sqrt{\pi\omega_0/\tau}$, the budget $d_P \leq \sqrt{\kappa}/10$ needs $T\Lambda \leq 0.14(\Lambda_*/\Lambda)$ from the force noise and $T\Lambda \leq 0.15(\Lambda_*/\Lambda)^2$ from memory: the pulses must outrun every retained field mode, and without a cutoff no $T > 0$ works (Fable referee). The resonant value $K = 1$ used for the $3\kappa/4$ example requires $\Lambda \ll \Lambda_*$; at $\Lambda \sim 1/\tau$, $K \sim 1/(\pi\tau\omega_0)$ and the gains must exceed $2K$.

Small independent Gaussian readout noises of standard deviations r_1, r_2 add at most r_1/G and Gr_2/B to the two budgets (16). Every chosen strict violation therefore permits positive readout noise. A prescribed minimum readout noise, coupling duration, gain ceiling or memory disturbance could restrict the protocol; those would be added instrument premises.

Order and cost of the limits. For a target area, first fix a finite cutoff and its actual prior V , then choose finite gains using (11), then choose a positive duration using (17). The constants depend on the cutoff and the gains. Removing the cutoff before this construction reintroduces the divergent full momentum variance. The impulsive limit also spends unbounded control strength and allows idealized bilinear couplings. This establishes the linear-model obstruction; relativistic locality, material bounds and an implementation using only electromagnetic controls would require further estimates with c . Heavy or detuned pointers are unnecessary for the theorem. Increasing mass can reduce coordinate noise or slow damping, but leaves the equilibrium resonant area κ unchanged.

4. A recording law that restores closure

There is a precise point where the two-pulse protocol can be blocked. After the first pulse, reading $Y + GQ$ and keeping pointer 1 available for the second pulse must itself have a conjugate cost.

An explicit repair. In the product resonant example, let the stored first reading have independent Gaussian error N of variance $s^2 > 0$, and let creating that reading give pointer 1 a further independent momentum kick D of variance d^2 . The body has already received $-G\Pi$; the second pointer now measures $\Pi + D$. Assume

$$s^2 d^2 \geq \kappa^2. \quad (20)$$

Grant arbitrarily accurate inference of $\Pi + D$ from pointer 2. The two independent sectors then give

$$\text{Var}(Q \mid R) = \frac{\kappa(\kappa + s^2)}{\kappa + s^2 + G^2\kappa}, \quad \text{Var}(P - G\Pi \mid R) = \kappa + \frac{G^2\kappa d^2}{\kappa + d^2}. \quad (21)$$

For $d^2 = \kappa^2/s^2$ their product is exactly κ^2 ; larger d^2 or finite precision of the second read increases it. The extra kick makes the later pointer momentum an imperfect record of the earlier body kick. A weaker statement that every read imparts some disturbance would leave (20) unproved.

The following sufficient premise handles arbitrary correlations and composition. Let Ω_n be the canonical Poisson matrix and impose on every Gaussian joint state, including reusable apparatus and memory,

$$V + i\kappa\Omega_n \succeq 0. \quad (22)$$

Admit symplectic transformations, independent ancillary states obeying (22), discarding subsystems, and Gaussian measurements on a subsystem that is then discarded. Such a measurement has independent Gaussian seed covariance Γ with $\Gamma + i\kappa\Omega_A \succeq 0$. Sharp single-coordinate measurements are obtained as squeezed limits. When a pointer is retained after being read, require a dilation of its reading within this same class; (20) is the basic coordinate example.

Proposition 3 (Gaussian closure). These operations preserve (22) on the unmeasured systems conditional on the complete record, including sequential adaptive uses. In particular the body's posterior has $\sqrt{\det \bar{\Sigma}} \geq \kappa$.

Proof. Symplectic transformations preserve (22), products give a direct sum, and discarding takes a principal submatrix. For a joint covariance $V = \begin{pmatrix} A & C \\ C^\top & B \end{pmatrix}$, Gaussian conditioning gives $A - C(B + \Gamma)^{-1}C^\top$. Complex conjugation of the seed condition gives $\Gamma - i\kappa\Omega_A \succeq 0$. Adding that positive matrix on the apparatus block of (22) gives

$$\begin{pmatrix} A + i\kappa\Omega_S & C \\ C^\top & B + \Gamma \end{pmatrix} \succeq 0.$$

Its Schur complement proves the desired conditional inequality. Limits preserve positivity. Apply this at every branch of a sequence, retaining all apparatus needed at later stages. $\square$

This is the Gaussian quantum instrument restriction written in classical covariance language. Bartlett, Rudolph and Spekkens derive Gaussian quantum preparations, transformations and measurements from an epistemic restriction (2012 paper, author abstract checked). Proposition 3 supplies the needed covariance proof here. The common-field covariance (8) fixes the prior, which already obeys (22); the seed restriction and the retained-pointer law are separate premises. Law (20) is the uncorrelated case; the dilation gives in general $\det \text{Cov}(N, D) \geq \kappa^2$. Even separate one-pair bounds on every prior leave the joint condition (22) to be established.

5. Constants, complementarity and the Newton comparison

The positive constant that a closure law would preserve is the resonant $\zeta = \kappa$. Under the additional SED/Planck identification in the revised link, §3 (local full-read),

$$\kappa = \frac{\hbar}{2} = \frac{h_P}{4\pi}, \quad h_* = 2\zeta = 2\kappa = \hbar = \frac{(\pi^4/15)^{1/3}}{2\pi} h_{\text{rad}}. \quad (23)$$

That thermal identification retains the conditions and disputed source status stated there. Theorem U supplies the radiation unit independently of any record restriction. Theorem B' in the fifth-postulate note, §4 (local full-read), classifies symplectically invariant, noise-closed one-pair Gaussian classes. Theorems 1 and 2 show why closure under *conditioning on retained records* is an additional requirement beyond that classification.

For Proposition 3's restricted Gaussian instruments and the affine generators of Theorem F, §4 (local full-read), Lemma G supplies statistical speed constant $h_* = 2\kappa$. Subject to Theorem F's remaining comparison hypotheses,

$$\frac{s}{8} \sum_j \sup \Delta D_j + \frac{J}{2} \sum_j \sup \Delta X_j \geq 2\kappa \arcsin(1 - 2\epsilon), \quad s = \frac{F\tau^2}{2m}, \quad J = F\tau. \quad (24)$$

The passive retained readings of Theorem 1 fall outside this restricted class. Their small posterior leaves the unconditional impulse spread charged in (24) in place. The unrestricted, non-Gaussian comparison requires the further operator/statistical-speed premises of Theorem C and Theorem F respectively.

Theorem E's window bound. Write $z_\epsilon = \Phi^{-1}(1 - \epsilon/2)$ for $0 < \epsilon < 1/2$, where Φ is the standard normal distribution function. For any conditional Gaussian, windows centred at its conditional means with half-widths $a = z_\epsilon \sigma_q$ and $b = z_\epsilon \sigma_p$ contain the respective variables with probability $1 - \epsilon$. The diagonal estimates in Theorem 1 give

$$ab \leq z_\epsilon^2 \left(\frac{\sigma_Y \sigma_P}{G} + \frac{\sigma_Y \sigma_{Z_2}}{B} \right) \rightarrow 0. \quad (25)$$

The canonical scaling (7) leaves this product unchanged. Thus the unrestricted classical record class can violate, at any supplied $\hbar > 0$, the necessary quantum condition

$$\lambda_0 \left(\frac{ab}{\hbar} \right) \geq (1 - 2\epsilon)^2$$

of Theorem E, §6b (local full-read). Equation (25) bounds the product of marginal spreads directly, so it also covers correlated posteriors whose determinant alone would give insufficient control of axis-aligned windows.

Under (22) with $\hbar = 2\kappa$, every Gaussian covariance has a Gaussian quantum-state realization: a symplectic transform of thermal oscillator states with covariance eigenvalues at least $\hbar/2$. Consequently Theorem E applies to those Gaussian marginals. Extending that statement to all non-Gaussian classical states would require an additional state/instrument principle.

Finally the parallel-record note, Proposition 1 (local full-read), gives the *forgotten* quantum record's momentum variance $\hbar^2/(4\sigma^2) = \kappa^2/\sigma^2$ under (23). Equation (20) requires this same cost when recording a pointer that will be reused. Forgetting a reading and preserving it for a later kick estimate impose different conditioning questions; Proposition 3 states a law that covers both.

6. Complete records: the continuity and noise mechanisms compared

This checkpoint uses the Gaussian instrument class of Proposition 3, the fixed-cutoff shared-bath model of Theorems 1–2, and Proposition L (local passage). Its new statements are exact finite-dimensional Gaussian calculations. Positivity, composition and radiation calibration are tested separately; no additional physical law is adopted.

6.1 Continuity must concern the complete apparatus family

Fix one body, $m > 0$, a window $[0, \tau]$ and a family $\mathfrak{A}$ of finite, non-adaptive affine apparatus protocols with Gaussian preparations and noises independent of the force hypothesis. Each protocol includes its entire retained memory, bath monitors, reused pointers and terminal readings in one record R_π . Linear dynamics gives

$$R_\pi = \mu_\pi + B_\pi \theta + F v_\pi + \xi_\pi, \quad \xi_\pi \sim N(0, V_\pi), \quad (26)$$

where θ comprises the unknown initial position and velocity. For preparation-ignorant tests take a full maximal invariant $Q_\pi R_\pi$, with $Q_\pi B_\pi = 0$. Put $b_\pi = Q_\pi v_\pi$, $W_\pi = Q_\pi V_\pi Q_\pi^\top$ and

$$I_\pi = b_\pi^\top W_\pi^\dagger b_\pi \quad \text{if } b_\pi \in \text{Ran } W_\pi; \quad I_\pi = \infty \text{ otherwise.} \quad (27)$$

The finite value is the force Fisher information. The singular case has a noiseless force-sensitive direction. The projection retains all invariant information, including correlations with side records; it is not a choice to discard inconvenient memory. With known preparation take $Q_\pi = I$ instead.

Proposition 4 (exact uniform-continuity criterion). In this family, for equal prior probabilities the best invariant error is

$$P_\pi(F) = \Phi(-|F|\sqrt{I_\pi}/2). \quad (28)$$

For $F \neq 0$ the $I_\pi = \infty$ value is zero; at $F = 0$ it is $1/2$. Writing $I_* = \sup_{\pi \in \mathfrak{A}} I_\pi$, the complete-record maximal-invariant laws $P_{\pi, F}$ are uniformly continuous at zero in total variation exactly when $I_* < \infty$. The quantifier covers every admitted schedule, number of marks and retained memory:

$$\sup_{\pi \in \mathfrak{A}} \|P_{\pi, F} - P_{\pi, 0}\|_{\text{TV}} = 2\Phi(|F|\sqrt{I_*}/2) - 1. \quad (29)$$

For $I_* = \infty$ the right side is one at every $F \neq 0$.

Proof. On the supported subspace whiten the Gaussian noise. The likelihood ratio depends only on the projection along the whitened mean difference, whose length is $|F|\sqrt{I_\pi}$. Thresholding at its midpoint gives (28) and the total variation in (29). Different affine supports in the singular case are disjoint. Monotonicity and continuity of Φ permit the supremum, whether or not attained. Consequently finite I_* gives the common modulus $|F|\sqrt{I_*}/\sqrt{2\pi}$, while infinite I_* gives a jump. $\square$

For the unrestricted independent-mark family of Proposition L, $I_* = \tau^3/(24m\kappa)$ at $\kappa > 0$. Its zero branch has $I_* = \infty$. Thus that proposition uses **uniform continuity over the family**, equivalently continuity of its optimal verdict, rather than pointwise continuity of finite noisy apparatus. A bounded classical family may have $I_* < \infty$ at $\kappa = 0$. For example three passive, independent position readings of variance σ^2 at $0, \tau/2, \tau$, with zero recoil, give the invariant statistic $R_0 - 2R_{\tau/2} + R_\tau$, mean $F\tau^2/(4m)$ and variance $6\sigma^2$. Its information is

$$I = \frac{\tau^4}{96m^2\sigma^2}.$$

More generally a finite total precision budget $\sum_j \delta_j^{-2} \leq C$ gives $I_* \leq C\tau^4/(4m^2)$ even with known preparation. Such budgets add under composition; maintaining the same C requires a resource restriction. Unlimited repeats on the same zero-recoil trajectory make $I_* \rightarrow \infty$.

Therefore (29) proves the exact statistical requirement of the continuity route. Supplying it independently for a physically specified family closed under arbitrary finite composition remains necessary. It alone supplies neither a terminal imprecision-kick law nor an action constant common to different m, τ and apparatus classes. The quantity $\tau^3/(24mI_*)$ has action units when $0 < I_* < \infty$; identifying it with a universal mark scale uses the additional mark-model saturation and universality premises.

6.2 The noise law must survive side observations of the kick

For any jointly Gaussian body X and complete record $\mathcal{R}$, the exact answer is the supported Schur complement

$$\text{Cov}(X | \mathcal{R}) = K_{XX} - K_{X\mathcal{R}} K_{\mathcal{R}\mathcal{R}}^\dagger K_{\mathcal{R}X}. \quad (30)$$

Appending a record decreases this covariance in the positive-semidefinite order. A prior noise covariance therefore needs a conditioning estimate, including indirect observations, before it can enforce a floor.

Here is an exact extension of (20)–(21). Start with the algebraic resonant example $V = \kappa I_6$, $\kappa > 0$, in canonical scaled coordinates. The first pulse still gives body $X = (Q, P - G\Pi)$ and pointer coordinate $Y + GQ$. Storing the latter produces error N and kicks the reusable pointer by D . Grant exact inference of its later momentum $K = \Pi + D$ from the second pointer, as in §4's limiting second readout. Let (N, D, S) be jointly Gaussian and independent of the six initial body–pointer variables; S is any side record of the readout noise. It can be a monitor read earlier or later, or stored in a separate memory. The complete record is

$$\mathcal{R} = (Y + GQ + N, K, S).$$

Subtract conditional means given S and put

$$\begin{aligned} E_S &= \text{Cov}((N, D)^\top | S) = \begin{pmatrix} a & c_{ND} \\ c_{ND} & b \end{pmatrix} \succeq 0, \\ x &= G^2, \quad u = \kappa + a, \quad v = \kappa + b, \\ \mathcal{D} &= (\kappa x + u)v - c_{ND}^2 > 0. \end{aligned} \quad (31)$$

The symbol c_{ND} is a covariance, distinct from the speed c . All of a, b, c_{ND}, u, v have action units.

Proposition 5 (the conditional recording law, necessary and sufficient in this model). The exact posterior determinant is

$$\boxed{\det \text{Cov}(X | \mathcal{R}) = \kappa^2 \left[1 + \frac{x(ab - c_{ND}^2 - \kappa^2)}{\mathcal{D}} \right]}. \quad (32)$$

For every nonzero gain G , the posterior area is at least κ if and only if $\det E_S \geq \kappa^2$. A finite-precision second reading can only increase the posterior covariance; the exact-reading limit gives the sharp necessity test.

Proof. Condition first on S . The remaining two readings have covariance $B_R = \begin{pmatrix} \kappa x + u & c_{ND} \\ c_{ND} & v \end{pmatrix}$, while $A_X = \text{diag}(\kappa, \kappa(1+x))$ and $C_{XR} = \text{diag}(G\kappa, -G\kappa)$. In particular (30) gives

$$\begin{aligned} \Sigma_{11} &= \kappa(uv - c_{ND}^2)/\mathcal{D}, \\ \Sigma_{12} &= -x\kappa^2 c_{ND}/\mathcal{D}, \\ \Sigma_{22} &= \kappa(1+x) - x\kappa^2(\kappa x + u)/\mathcal{D}. \end{aligned} \quad (33)$$

For a compact determinant check, the covariance of the readings conditional on X is $\begin{pmatrix} u & c_{ND} \\ c_{ND} & b + \kappa/(1+x) \end{pmatrix}$. Taking the joint determinant in both block orders yields

$$\det \Sigma = \kappa^2 \frac{\kappa u + (1+x)(ub - c_{ND}^2)}{(\kappa x + u)v - c_{ND}^2}.$$

The numerator minus the denominator is $x(ab - c_{ND}^2 - \kappa^2)$, proving (32). The initial strictly positive κI_6 makes the reading covariance invertible, including singular E_S . $\square$

A concrete indirect-monitor test. Let N, D be independent with $\text{Var } N = a_0 > 0$, $\text{Var } D = b_0 > 0$ and $a_0 b_0 = \kappa^2$, so the marginal recording law (20) is saturated. Allow the side record $S = D + M$, where M is an independent Gaussian monitor error of variance $r^2 > 0$. Then

$$a = a_0, \quad b = \frac{b_0 r^2}{b_0 + r^2}, \quad c_{ND} = 0, \quad \det E_S = \kappa^2 \frac{r^2}{b_0 + r^2} < \kappa^2. \quad (34)$$

Every informative monitor, even a noisy one, makes (32) strictly smaller than κ^2 for $G \neq 0$. At $r \downarrow 0$, the kick is known and $\det \Sigma = \kappa^2 u / (u + \kappa G^2) \rightarrow 0$ as $G \rightarrow \infty$. At $r \rightarrow \infty$ the original saturated repair is recovered. All limits keep their order and apparatus assumptions explicit. A realizable monitor may itself disturb the pointer; the passive monitor in this test is a granted operation whose physical admissibility is exactly what a closure mechanism must settle.

For arbitrary correlated readout noise with covariance E_0 and side record covariance V_S , the matrix to test is

$$E_S = E_0 - \text{Cov}((N, D), S) V_S^\dagger \text{Cov}(S, (N, D)). \quad (35)$$

The common bath's spectrum can set E_0 , but it leaves this conditional innovation to be controlled. In the general correlated prior of (8), use (30) on the full joint record; (32)'s iff is asserted for its stated product-prior example only. Theorems 1–2 already cover arbitrary finite body–pointer cross correlations. Proposition 3 supplies a sufficient composition law on every retained subsystem; (34) identifies a side observation that must be priced or excluded by that law.

6.3 Ideas supplied by the current Lean formalization

The sister repository is read at commit [381d5e9](#), keeping its historical theorems and modern diagnostics separate. Its dependency graph and Proposition I realization obligations (local full-read) expose a specific missing bridge in both printed stages: the cited Lemma III Cor. 4 concerns approximating a given curve, while the force-generated vertices move under refinement. The finite area law (P1) and refinement identities (P2) are checked; realization (P3) and force identification (P5) remain separate. The harmonic field has a mesh-uniform quadratic stability invariant. The pending norm inequality in its Fraction representation is a formalization obstacle; a weighted sum norm is a candidate for the finite stability/defect argument. It is not evidence that classical refinement requires an action floor.

The graph suggests the same distinction for records: geometric control of the body and apparatus map needs an additional statistical control in the complete-record norm. The following finite declarations were read in their definitions and proofs (local full-read). Their algebra supplies two concrete tests for that statistical bridge. The Gaussian consequences here are written derivations, separate from the compiled finite theorems.

First, `RelativeMotion.corVI_relative` and `relative_deflection_difference` derive exact cancellation of an arbitrary common deflection history from both relative coordinates. A classical Gaussian consequence is immediate: if two observed motions receive the same acceleration history, differencing them removes that random history too. More generally if a retained signal is $r + L\eta$, with bath covariance C_η , a difference map D_r leaves covariance

$$D_r L C_\eta L^\top D_r^\top. \quad (36)$$

If $D_r L = 0$ it is zero regardless of the bath amplitude. This is a modern noise-model consequence of the finite cancellation identity, not a historical assertion about radiation. A noise floor must survive accessible relative observations as well as the conditional subtraction (35). Common driving alone does not guarantee that survival.

Second, `PartitionComparison.partition_gap` and `PartialCell.candidate_partial_residual` and the partial-cell proof separate the actual end-kick polygon from its rational constant-force

comparison map. At any rational sample time $t \leq \tau$ the exact position residual is $a_{\text{acc}}c_\pi(t)$, where

$$c_\pi(t) = \frac{1}{2} \left(\sum_{j \text{ complete}} h_j^2 + u^2 \right), \quad 0 \leq c_\pi(t) \leq \frac{1}{2}t|\pi|, \quad a_{\text{acc}} = F/m. \quad (37)$$

Here u is the partial final duration and $|\pi|$ bounds every complete and partial cell. The formal theorem uses represented fractions and cross-multiplication equivalence; (37) is its rational-value reading. It fixes the force signal's geometric discrepancy uniformly over partitions. It supplies no measurement law.

For a single Gaussian position record of width σ_π the total variation between polygon and corrected-map records is exactly

$$2\Phi \left(\frac{|F|c_\pi(t)}{2m\sigma_\pi} \right) - 1. \quad (38)$$

Thus the mechanically vanishing residual gives statistically convergent records only when its noise-normalized size vanishes. A width $\sigma_\pi = F_0 c_\pi(t)/m$ with fixed force $F_0 > 0$ leaves (38) constant for every nonzero F , even as the mesh vanishes. A fixed positive width makes it tend to zero. For the complete multivariate record the corresponding condition is $\|W_\pi^{-1/2}Q_\pi(v_\pi - v_{\text{cont}})\| \rightarrow 0$ on its supported subspace; a noiseless residual direction again has total variation one. This distinguishes uniform mechanical refinement from uniform complete-record refinement by an explicit calculation.

The same repository's `PhaseArea.cells_area2` preserves every phase triangle under affine drift/kick schedules. It supports the action *kind* of a symplectic covariance restriction, while preserving triangles of every initial size. Its invariant gives no selected minimum. The end-kick convention of (37) differs from the symmetric half-kick construction of the insertion note; each statement retains its own convention. These tests concern records and the polygon–trajectory discrepancy, rather than Kepler sectors.

6.4 Mechanism selected and the next lemma

The noise route now has a sharper constructive target: a body–apparatus model must retain a **conditional innovation pair** with determinant at least κ^2 after every admissible side record. Its readout dynamics must charge observations of the kick and of reusable memory, as (34) shows. The immediate lemma is a branchwise innovation estimate for finite adaptive linear recording, followed by independence from the refinement schedule. Reject a proposed common-bath mechanism if an admissible relative channel annihilates its noise as in (36), or if a terminal side record reduces its innovation determinant below κ^2 as in (34).

The continuity route has been reduced to (29), but no independently derived uniform bound over a composition-closed physical family is available. Neither route has discharged positivity from independent physical premises. We therefore select the conditional-innovation calculation as this checkpoint's advance, and the innovation law as the next mechanism to construct. No new physical premise has been installed. Within an admitted Gaussian closure law $h_* = 2\kappa > 0$; proving that law, proving its positive scale and identifying $h_* = \gamma h_{\text{rad}}$ remain distinct. Formula (23)'s radiation calibration retains its existing conditional source status.

7. Consequence for STATE

The Newton task advances from testing stationary radiation to deriving an admissible recording law. Shared Gaussian driving, including its cross correlations and radiation damping, allows the two-pointer counterexample at every fixed finite cutoff when classical retained readouts and the stated controls are available. The next physical premise to justify is a restriction such as (20)–(22) for reusable pointers and their memories. The referee's analysis narrows it: the background's prior already obeys (22), so the premise is a bound on every terminal classical

readout, a restriction on observation equivalent at $\kappa = \hbar/2$ to Gaussian quantum measurement theory; an electrodynamic instrument constraint would only relocate it. The conditional scale identification and the unregulated momentum problem retain the scope stated in the revised SED link. The September 29 checkpoint sharpens the recording law to conditional innovation (32)–(35), quantifies uniform continuity by (29), and imports the Lean common-mode and noise-normalized refinement tests (36)–(38). Next alternate to the gauge connector expansion, then test adaptive innovation and complete retained memory in this model.